

Progress Report

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July 2026

1 Basic Project Description

Epilepsy diagnosis relies on manually reviewing hours of continuous EEG. This project builds a faster, automated detection system to support neurologists. Given a short raw EEG segment, the model outputs a seizure/non-seizure classification and a Grad-CAM visualization of the driving region (Figure 1). Deep learning is well suited here since relevant EEG patterns exist jointly in frequency, learned by the CNN, and its evolution over time, learned by the LSTM. While CNN-LSTM architectures have been applied to this dataset before, this project’s contribution is evaluating how detection performance scales with the number of training patients, motivated by evidence that many existing studies do not rigorously separate patients across training and evaluation.



Figure 1: CNN-LSTM architecture used for preliminary seizure detection.

2 Data Processing

This project uses the CHB-MIT Scalp EEG Database which contains EEG recordings from pediatric patients. Patient chb01 is being used for training and chb02 was used to test generalization on unseen data. The full 24-patient dataset will be used for the final report. Each raw recording is bandpass filtered between 0.5 and 40 Hz followed by common average referencing. The filtered recordings are segmented into 4 second windows that don’t overlap. Each window is labelled as seizure if it overlaps a documented seizure interval by at least one second. Each window is then converted into a log-magnitude spectrogram with short-time Fourier transform. It retains frequencies up to 40 Hz to match the bandpass filtering already applied. Across the seven seizure-containing record-

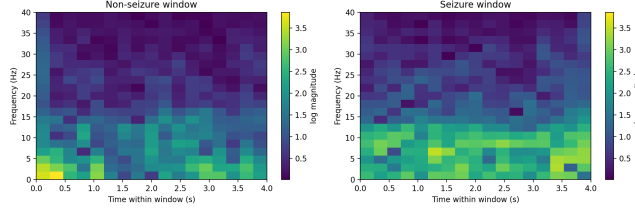


Figure 2: Example of a cleaned training sample.

ings for patient chb01, only 1.9% were named seizure. This illustrates the severe class imbalance inherent to this task.

Handling this data also required accounting for file numbering that was not contiguous, requiring the building of file lists from what actually exists on disk rather than an assumed naming pattern.

3 Baseline Model

As a baseline, a hand-coded amplitude detector was implemented. For each patient, a mean and standard deviation of absolute EEG amplitude were computed from known seizure periods. A window was flagged as seizure if at least two channels exceeded a threshold of mean plus 2.5 standard deviations for at least two continuous seconds. As shown in Figure 3 true and false positives both can produce amplitude excursions above the threshold. This illustrates why amplitude alone is an insufficient signal, motivating the use of a learned model.

Table 1: Quantitative baseline amplitude-threshold detector performance on the held-out `chb01_04.edf` file.

Metric	Value
Sensitivity	0.5000
Specificity	0.9271
Precision	0.0580
F1 score	0.1039
Accuracy	0.9233

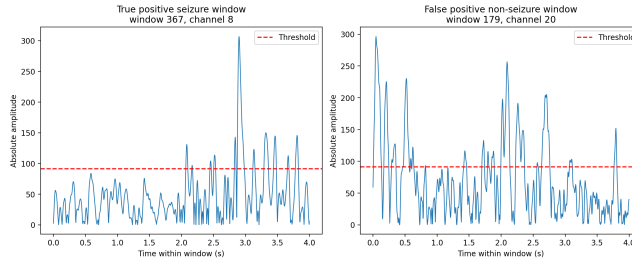


Figure 3: A qualitative comparison of one correctly detected seizure window against one falsely flagged non-seizure window

4 Primary Model

As a proof-of-concept toward the proposed ResNet-18 and LSTM architecture, a lighter-weight CNN-LSTM was trained end-to-end to validate the full pipeline. Evaluated on a held-out recording from the training patient, the model correctly identified every sequence in that file’s seizure event; evaluated without retraining on an unseen patient, it again identified every seizure sequence, but specificity dropped substantially. This pattern, perfect sensitivity with collapsing specificity on an unseen patient, suggests the model recognizes seizure-like features but not yet patient-general background variation, motivating the planned scaling experiment across training patients. With only 12 seizure sequences in training, a weighted sampler was used to counteract class imbalance, so results should be read cautiously.

Table 2: Sensitivity, specificity, precision, and F1 for the trained model evaluated within-patient versus on an unseen patient, showing perfect sensitivity in both cases but a sharp drop in specificity cross-patient.

Metric	Within-patient	Cross-patient
Sensitivity	1.0000	1.0000
Specificity	0.9538	0.4579
Precision	0.1961	0.1655
F1 score	0.3279	0.2840
Accuracy	0.9543	0.5105

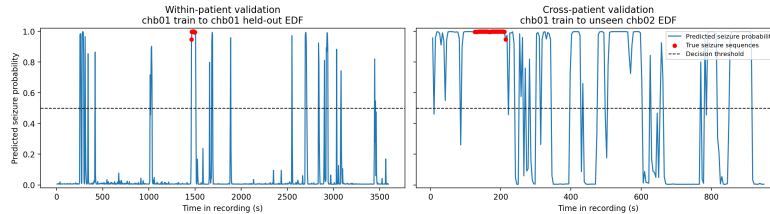


Figure 4: Predicted seizure probability over time for the within-patient and cross-patient validation recordings, with true seizure events and the decision threshold marked, illustrating the rise in false positives on the unseen patient.

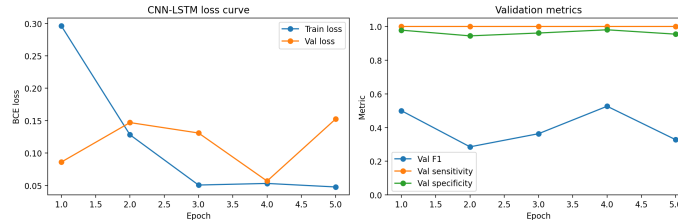


Figure 5: Training and validation loss alongside validation F1, sensitivity, and specificity across 5 epochs, with final-epoch values reported to avoid selecting the best-performing epoch.

References

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